

Mass-Anchored Glagov-Constrained CT-FFR

A Staged Reduced-Order Development Framework in the DISCHARGE Study

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1 Abstract

Background: Many reduced-order CT-FFR pipelines use lumen-only geometry and diameter-based flow scaling. This creates two distinct risks: reference vessel underestimation in positively remodelled arteries and potentially miscalibrated flow allocation when myocardial demand is inferred from vessel caliber alone.

Objective: We aim to develop a technically coherent, staged CT-FFR framework for DISCHARGE in which invasive FFR validation is the primary development endpoint and prognostic analysis is a secondary extension.

Methods: We define four pre-specified model variants in a 2×2 design: lumen-only vs Glagov-constrained reference geometry, and diameter-based vs mass-based flow allocation. The reduced-order pressure-drop model uses actual lumen area in the distributed viscous term and reference area at the MLA site in the reference-dependent contraction term. For mass-based flow, the primary formulation uses vessel-territory myocardial mass scaling with allometric sensitivity analysis spanning $\gamma \in [0.75, 1.0]$. Existing Medis vessel segmentations are treated as partial seed data requiring harmonization, centerline completion, and quality control before flow derivation.

Primary endpoint: Diagnostic validation against the invasive FFR subset available in DISCHARGE, with up to 113 recorded cases before matching and quality control and a conservative expectation of approximately 70–90 analyzable paired target-vessel comparisons after geometry QC.

Secondary endpoint: Segmentation robustness and inter-method agreement.

Exploratory endpoint: Patient-level prognostic association with downstream outcomes in a larger harmonized CT subset.

Conclusion: The DISCHARGE CT-FFR program should be framed as a staged method-development study rather than as an immediate full-cohort prognostic analysis. The key technical correction is explicit: Glagov information enters the reference-dependent contraction term, whereas the viscous term remains lumen-based.

2 Introduction

2.1 Scientific premise

Invasive fractional flow reserve is an accepted physiological standard for lesion-level assessment and revascularisation guidance, grounded in foundational validation and strategy trials^{1–3}. CT-FFR ex-

tends this standard non-invasively. The biological rationale for improving CT-FFR reference geometry is well established: coronary arteries remodel outward during early atherosclerosis, preserving lumen caliber despite substantial plaque burden^{4–6}. Lumen-only methods therefore risk underestimating reference size in positively remodelled lesions. Separately, vessel caliber is an imperfect surrogate for demand; flow is driven by the perfused myocardial territory, not by vessel caliber alone^{7–9}.

The project hypothesis is therefore split into two testable components:

1. **Reference hypothesis:** Outer-wall-constrained reference reconstruction improves pressure-drop estimation relative to lumen-only reference assumptions.
2. **Demand hypothesis:** Vessel-territory mass scaling improves flow allocation relative to diameter-based scaling.

The value of the project lies in testing these two hypotheses independently and jointly.

2.2 The Glagov bias and why it matters

Outward remodeling creates a specific reference-geometry problem. When plaque burden is partly accommodated by expansion of the external vessel boundary, the observed lumen may appear only mildly narrowed even though the healthy reference vessel would have been substantially larger. A lumen-only reference can therefore understate stenosis severity and reference-dependent contraction loss in precisely the lesions where anatomy and physiology are most likely to diverge. We use **Glagov bias** as shorthand for this systematic underestimation of healthy reference caliber in positively remodelled arteries.

Positive remodeling is a well-recognized coronary stenosis pattern: IVUS studies demonstrate that compensatory expansion of the external elastic membrane (EEM) can coexist with substantial plaque burden while lumen area remains near-normal^{10,11}. This is precisely the anatomical setting where lumen-only CT-FFR reference geometry is most likely to underestimate true stenosis severity.

2.2.1 Didactic numerical example

Parameter	Lumen-envelope reference	Glagov-constrained reference
Observed lumen at MLA	4.0 mm ²	4.0 mm ²
Inferred reference area	5.0 mm ²	7.5 mm ²
Calculated area stenosis	20%	47%
Illustrative CT-FFR	0.92	0.82
Clinical interpretation	Reassuring	Borderline / warrants closer review

These values are intentionally schematic. Their purpose is to show how identical observed throat geometry can imply very different reference areas, percent stenosis, and reduced-order contraction losses once compensatory remodeling is acknowledged. They are not projected performance results and should not be read as the exact behavior of the primary DISCHARGE implementation.

2.3 Prior art and why a transparent reduced-order framework is still needed

The clinical value of CT-FFR is supported by early validation studies, later registry experience, and randomized testing-strategy evidence, including PRECISE^{12–15}. The unresolved issue is not whether CT-FFR is useful in general, but whether its reference geometry and boundary-condition assumptions remain adequate in remodelled arteries and altered-demand states.

Current method families each leave that question partly open. Full 3D CFD pipelines are physiologically rich but computationally expensive and often opaque in implementation¹⁶. Fast reduced-order analytic approaches, including those evaluated in the FAVOR studies, show that pressure-drop models can be clinically credible at much lower runtime, but they still rely primarily on lumen-derived reference assumptions^{17,18}. Machine-learning surrogates reduce runtime further, yet they generally inherit the geometric representation encoded in the training data and offer less direct control over how reference size and flow demand enter the model^{19,20}.

The gap is most clearly seen by comparing method families across the two corrections this project targets:

Approach	Reference geometry	Flow boundary condition	Transparency	Runtime	Gap addressed here
3D CFD (HeartFlow FFRCTA, Siemens cFFR; ¹⁶)	Lumen-derived	Diameter-based scaling	Low	Hours	Neither
Reduced-order analytic (FAVOR II ¹⁷)	Lumen-derived	Diameter-based scaling	High	Seconds	Neither
ML surrogate (^{19,20})	Inherited from training data	Diameter-based or fixed	Low	Seconds	Neither
Outer-wall-informed invasive imaging	Outer-wall constrained	Fixed or diameter-based	Moderate	Hours	Reference only

Approach	Reference geometry	Flow boundary condition	Transparency	Runtime	Gap addressed here
Present: MG-CT-FFR	Glagov-constrained (CT outer wall)	Territory mass-based	High	Seconds	Both

A transparent reduced-order CT-FFR framework that combines outer-wall-constrained reference geometry with territory-mass flow allocation has not, to our knowledge, been published in the current invasive-validation literature.

A relevant recent prior art example is²¹, who apply a distal-vessel CT-FFR framework for serial monitoring of epicardial disease. That work demonstrates the potential of CT-derived physiology beyond single-point diagnosis, but like its predecessors it derives the reference vessel from the lumen itself and uses diameter-based flow boundary conditions rather than myocardial mass allocation. These are the two assumptions the present framework is designed to test directly.

The present framework is therefore not a rejection of prior CT-FFR work. It is a targeted attempt to test two specific corrections inside a transparent reduced-order setting: reference reconstruction in remodelled arteries and demand allocation from myocardial mass rather than vessel caliber alone.

2.4 Why DISCHARGE is the right setting

DISCHARGE is distinctive not because every vessel is already flow-ready, but because a single program can link partial vessel contours, repeated reads, an invasive FFR subset, and broader CT follow-up within a harmonized clinical ecosystem^{22,23}. That combination remains unusually strong for a staged method-development study, provided the geometry burden and analyzability limits are described conservatively.

2.5 Study scope and structure

This manuscript is framed as a staged method-development study with four operating principles:

1. Invasive FFR validation is the primary development endpoint.
2. Segmentation availability and cohort readiness are described conservatively.
3. The mathematical role of Glagov reference geometry is made explicit in the pressure-drop model.
4. Mass-based flow is defined at the vessel-territory level rather than by assigning total LV demand to an individual vessel.

The remainder of the manuscript follows that logic directly: data reality and staged cohorts first, then the 2×2 model family, segmentation and geometry workflow, Glagov reference reconstruction, reduced-order hemodynamics, flow allocation, validation, implementation roadmap, and discussion.

3 Data Reality and Cohort Strategy

3.1 What is available now

The DISCHARGE ecosystem remains uniquely valuable, but the usable vessel dataset is not a fully harmonized full-tree resource.

Current starting assets:

- partial Medis lumen and outer-wall vessel segmentations across a large CT cohort;
- segment-level point-cloud exports rather than fully standardized coronary trees;
- a subset with dual-reader vessel data for reproducibility work;
- an invasive FFR subset of 113 cases before image matching, lesion mapping, and geometry quality control;
- broader CT clinical data and follow-up that may support later exploratory prognostic analyses.

These assets can be summarized more explicitly as follows:

Asset	Current operational status	Immediate role in the staged programme
Medis lumen and outer-wall vessel exports	Available across a large CT subset, but partial and not yet harmonized as full coronary trees	Seed geometry for Cohorts A and B
Segment-level contour / point-cloud vessel exports	Available	Reconstruction of ordered lumen and outer-wall area profiles
Dual-reader or repeated vessel reads	Available in a subset only	Reproducibility and reader-agreement analyses
Invasive FFR-linked cases	113 recorded before CTA matching, lesion mapping, and strict geometry QC	Source population for Cohort A

Asset	Current operational status	Immediate role in the staged programme
Broader CT clinical and follow-up data	Available, but not yet tied to a harmonized flow-ready vessel dataset	Future Cohort C exploratory analyses
CT-derived LV mass workflow	A candidate workflow exists, but it is not yet treated as a locked cohort-wide quantitative input	Supports later mass-based model expansion once calibrated and QC-locked

3.2 What is not yet available

The project should not claim:

- complete outer-wall coverage for all major vessels in all CT cases;
- immediate full-tree readiness for reduced-order flow computation;
- immediate full-cohort prognostic processing without additional segmentation harmonization.

Instead, additional work is required to achieve:

- continuous proximal-to-distal vessel coverage in analyzable target vessels;
- consistent centerline and slice ordering across cases;
- lesion-to-FFR mapping;
- outer-wall plausibility control in calcified and noisy segments;
- enough geometric completeness to support per-vessel pressure-drop computation.

This does not mean the project becomes unusable whenever outer-wall segmentation is imperfect. A central contingency principle is that reference modeling should be hierarchical rather than binary: use observed outer-wall information where a trusted interval supports Glagov reconstruction, revert locally to the lumen-envelope comparator where it does not, and reserve stronger prior-based reference-lumen reconstruction for explicitly labeled sensitivity analyses. The practical strength of this design is that incomplete outer-wall availability becomes a structured modeling problem with reportable operating modes rather than a hidden source of overclaiming.

3.3 Staged cohorts

We therefore define three nested cohorts:

Cohort	Purpose	Role
A. FFR-linked development cohort	Primary model development and validation	Approximately 70–90 analyzable paired target-vessel comparisons from the 113-case invasive FFR subset after CT matching, lesion mapping, and geometry QC
B. Geometry/robustness cohort	Glagov and segmentation-method development	Cases with best outer-wall coverage, including repeat reads and dual-tool segmentations where available
C. Exploratory prognostic cohort	Secondary patient-level clinical testing	Larger harmonized CT subset after vessel processing has stabilized

This staging is more credible than assuming that all CT-arm cases can immediately support flow derivation.

4 Model Family

4.1 Four pre-specified model variants

The technical comparison is organized as a 2×2 ablation:

Geometry	Flow model	Name	Role
Lumen-only reference	Diameter-based	L-CT-FFR	Baseline comparator
Lumen-only reference	Mass-based	M-CT-FFR	Isolates demand effect
Glagov reference	Diameter-based	G-CT-FFR	Isolates reference effect
Glagov reference	Mass-based	MG-CT-FFR	Full model

This is preferable to a three-model hierarchy because it attributes improvement unambiguously. The design also allows combined benefit to be estimated directly: MG-CT-FFR may outperform the single-component arms, but interaction is treated as an empirical question rather than assumed. A pre-specified repeated-measures regression with geometry, flow-allocation, and their interaction as terms will estimate additivity versus non-additivity directly; in the current revision cycle, that interaction analysis is treated as a supportive secondary inferential analysis rather than as part of the primary multiplicity family.

4.2 Primary and secondary questions

Primary diagnostic questions:

1. Does G-CT-FFR outperform L-CT-FFR in the FFR-linked cohort?
2. Does M-CT-FFR outperform L-CT-FFR?
3. Does MG-CT-FFR outperform L-CT-FFR?

Secondary technical questions:

1. How sensitive are model outputs to outer-wall uncertainty?
2. Is the geometry-by-demand interaction directionally consistent with additive or non-additive combined benefit?
3. How stable are results to territory-fraction assumptions?
4. How large is the difference between linear territory-mass scaling and allometric scaling?

If one model is advanced beyond the internal 2×2 development comparison, the pre-specified ranking hierarchy is: highest AUC first, then lowest mean absolute error, then calibration closest to the identity line.

5 Segmentation and Geometry Workflow

5.1 Design principle

The development target is not a complete coronary tree for every subject. For stage-1 validation, the geometric target is a **continuous target-vessel segment** extending:

- from a proximal reference region;
- across the lesion region;
- to a distal lumen segment beyond the lesion and, where sensor-location metadata are available, approximating the invasive FFR sensor location.

This span is sufficient to compute CT-FFR in the matched vessel for development-phase validation, provided outlet demand and side-branch losses are handled by the prespecified reduced-order model.

This scope should be distinguished from two broader goals outside it. It is not equivalent to **full-tree CT-FFR**, which requires a substantially more complete reconstruction of coronary topology and branch geometry. Nor is it the same as a **lesion-only** or **ΔFFR-style** analysis, which targets the local translesional pressure drop rather than an absolute distal FFR estimate in the matched vessel.

5.2 Processing steps

1. Match CT case, vessel export, and invasive FFR metadata.
2. Harmonize slice ordering and centerline direction.
3. Convert cross-sectional contours into ordered area profiles $A_{\text{lum}}(s)$ and, where available, $A_{\text{outer}}(s)$.
4. Identify continuous vessel spans suitable for flow derivation.
5. Flag and exclude cases with insufficient geometric continuity, severe calcific blooming, or implausible outer-wall contours.

5.3 Segmentation work packages

The segmentation burden is part of the project, not a solved prerequisite:

- lumen harmonization and continuity checking;
- outer-wall completion or correction where Medis coverage is partial;
- quality control of calcified and ambiguous segments;
- centerline generation and resampling;
- reproducibility studies on repeated segmentations and tool comparisons.

6 Shared notation

The manuscript, proposal, and segmentation reference use the same core symbols:

Symbol	Meaning
$A_{\text{lum}}(s)$	observed lumen cross-sectional area along vessel arc length
$A_{\text{outer}}(s)$	outer-wall cross-sectional area where available
$A_{\text{ref}}(s)$	reconstructed local reference area profile, either lumen-envelope or Glagov-based
s	proximal-to-distal vessel arc length
M_{cal}	calibrated CT-derived LV myocardial mass entering the flow model
$M_{\text{ref,cal}}$	reference calibrated LV myocardial mass used to normalize the allometric sensitivity model (primary anchor 250 g; defined explicitly below)
f_v	vessel territory fraction assigned to vessel v
q_{rest}	resting perfusion per gram, typically handled as 1.0 mL/min/g $\approx 1.67 \times 10^{-8}$ m ³ /s/g when converted to SI units
H	hyperemic flow multiplier

In `segment_v1.md`, the same quantities appear in discrete indexed form such as $A_{\text{lum},i}$, $A_{\text{outer},i}$, $A_{\text{ref},i}$, and s_i .

7 Glagov-Constrained Reference Geometry

7.1 Reference reconstruction

For target vessel arc-length position s :

$$r_{\text{ref,G}}(s) = r_{\text{outer}}(s) - T_{\text{healthy}}$$

where T_{healthy} is a single vessel-level scalar estimate applied uniformly along the analyzed span, with

$$r_{\text{ref,G}}(s) \geq 1.05 r_{\text{lum}}(s)$$

and monotonic smoothing to enforce physiological taper.

In downstream area form, this radius safeguard implies $A_{\text{ref,G}}(s) \geq 1.05^2 A_{\text{lum}}(s) = 1.1025 A_{\text{lum}}(s)$, so the implied area floor is larger than the 5% radius margin from which it derives, a natural consequence of the quadratic radius-to-area relationship.

7.2 Healthy wall-thickness estimation

Primary rule:

$$T_{\text{healthy}} = \text{median}(r_{\text{outer}} - r_{\text{lum}})$$

measured over segments meeting all of the following:

- low local lumen-gradient;
- no obvious plaque or calcific blooming at the candidate reference site;
- adequate contour quality;
- minimum contiguous length threshold.

All four criteria must be satisfied. If no qualifying contiguous span meeting the minimum-length threshold remains after these exclusions, the fallback rule applies.

Fallback rule:

$$T_{\text{healthy}} = \beta \tilde{r}_{\text{lum}}, \quad \beta \approx 0.18$$

with $\beta \approx 0.18$ treated as an empirical prior anchored to coronary wall-thickness literature^{24–26}, and with clipping to a total-wall prior range of **0.40–0.80 mm**.

This range is deliberately wider than the 0.30–0.60 mm clip used in some intima-media-thickness references, because the CT-visible outer-wall contour reflects total wall thickness (intima + media + adventitia), not intima-media thickness alone. The lower bound 0.40 mm is physiologically conservative for a total-wall estimate; the upper bound 0.80 mm accommodates moderate calcified adventitia contribution while preventing implausible outliers. IVUS wall-thickness studies therefore serve as physiological anchors rather than as a one-to-one mapping of an intima-media measurement onto the CTA-visible total-wall contour^{24–26}.

When outer-wall delineation is locally unreliable, the project should not pretend that Glagov information is still observed. The primary action is local reversion to the lumen-envelope comparator. As a secondary sensitivity-only fallback, one may estimate a hypothetical healthy lumen radius from the

observed lumen profile using robust kernel-smoothed reference reconstruction. Blanco et al. describe such a weighted Gaussian-kernel healthy-radius estimator in the supplementary material of their 1D-versus-3D FFR study, building on earlier CTA stenosis-detection work by Shahzad et al.^{27,28}. In the present project, that type of method should be interpreted as a **reference-lumen fallback**, not as recovered outer-wall anatomy and not as primary Glagov evidence.

This contingency structure is scientifically useful rather than merely defensive. It means the project is not forced into the unrealistic claim that reliable outer-wall contours must exist everywhere before any meaningful development can begin. Instead, cases can be stratified into explicit operating regimes: direct Glagov mode when local wall data are trustworthy, lumen-envelope mode when they are not, and prior-supported reference-lumen sensitivity analyses when one wants to test how much information could still be recovered from lumen geometry alone. That preserves methodological honesty while reducing dependence on universal outer-wall success.

7.3 Hierarchical reference-reconstruction fallback modes

The fallback problem should be defined on the reference geometry $r_{\text{ref}}(s)$ rather than on an unrealistic requirement to recover the true outer wall everywhere. We therefore distinguish four operating levels:

- **Level 0:** measured outer wall over a trusted MLA-centered lesion/reference interval; this is standard Glagov mode and the primary evidence level for G-CT-FFR and MG-CT-FFR.
- **Level 1:** short-gap local reference reconstruction from trusted proximal and distal outer-wall anchors along the centerline, for example shape-preserving interpolation or spline interpolation.
- **Level 2:** physiology-constrained or lumen-derived reference-lumen fallback when direct wall anchors are inadequate, for example monotone taper enforcement, Murray-style branch-order scaling²⁹, or kernel-smoothed healthy-radius reconstruction^{27,28}.
- **Level 3:** population or anatomical priors only; these may be useful as sensitivity analyses or exclusion-grade comparators, but they should not be treated as primary Glagov evidence.

None of these levels changes the observed lumen profile $A_{\text{lum}}(s)$ used in the viscous term. They only change the support available for $A_{\text{ref,MLA}}$ or related reference-dependent sensitivity analyses. If the MLA neighborhood is supported only by Level-3 priors, the vessel is excluded from primary G-CT-FFR and MG-CT-FFR analyses and either reverted to the lumen-envelope comparator or retained only for sensitivity analyses. Levels 1 and 2 are methodologically useful contingencies, but they remain secondary support rather than primary Glagov evidence.

7.4 Exploratory remodeling biomarker: RI_{Glagov}

If the MLA neighborhood is supported by direct Level-0 wall data, the same reconstruction can also yield an exploratory remodeling index

$$\text{RI}_{\text{Glagov}} = \frac{A_{\text{outer}}(s_{\text{MLA}})}{A_{\text{ref,G}}(s_{\text{MLA}})}.$$

This differs from the classical lesion/reference remodeling index by normalizing the lesion outer-wall area to the model-derived local reference rather than to a nearby segment that may itself be diseased. In diffuse disease this is conceptually useful because it reduces dependence on choosing an adjacent visibly mild segment as the comparator, although $A_{\text{ref,G}}(s_{\text{MLA}})$ still depends on the local wall-thickness estimator, which itself requires a proximal or distal trusted reference span, and therefore does not eliminate dependence on nearby reference support entirely.

RI_Glagov is not a primary endpoint of the manuscript. In the current revision cycle it should be treated as a narrow exploratory **outer-wall-to-reference-lumen ratio** and reported only when the MLA neighborhood has direct Level-0 measured-wall support; otherwise it should be exported as **in-determinate** (Appendix A5.6). Because the current denominator is the model-derived local reference lumen area rather than an expected outer-wall reference quantity, the ratio should not be assigned positive/intermediate/constrictive bins in the current manuscript.

7.5 Lumen-only reference comparator

For the lumen-only models, the reference is derived from a proximal lumen envelope:

$$r_{\text{ref,L}}(s) = \text{monotone envelope of } r_{\text{lum}}(s)$$

This preserves a fair comparator without invoking outer-wall information.

8 Reduced-Order Hemodynamic Model

8.1 Key correction

The critical physical distinction is:

- **actual lumen area** determines the viscous resistance because blood flows through the lumen that exists;
- **reference area** determines the contraction/separation loss because this term quantifies how abruptly flow contracts relative to the healthy reference vessel.

This is the key modeling point: Glagov reference geometry must enter the pressure-drop model at the contraction term. Placing it in the viscous term would be physically incorrect, because blood flows through the lumen that exists, not through a reconstructed reference lumen.

8.2 Viscous term

We retain the distributed lumen-based viscous formulation:

$$\Delta P_{\text{visc}} = K_v \mu Q \int_0^L \frac{ds}{A_{\text{lum}}(s)^2}$$

with adopted reduced-order coefficient $K_v = 32$, $\mu = 0.004$ Pa s, and lumen area measured directly from cross-sectional contours³⁰. Here $K_v = 32$ is used as the primary Young-Tsai-style reduced-order convention, not as the exact circular-pipe coefficient. Appendix A2 shows that the exact circular area-form coefficient is $8\pi \approx 25.13$, so sensitivity analyses bracket the primary choice over $[8\pi, 32]$.¹⁸ demonstrate that an additional turbulence term improves accuracy beyond the two-term form at intermediate-to-severe stenoses; this extension is noted in Appendix A2 and deferred to post-validation development.

8.3 Reference-dependent contraction term

The Glagov correction enters the contraction term:

$$\Delta P_{\text{sep}} = K_i \frac{\rho}{2} Q^2 \left(\frac{1}{A_{\text{MLA}}} - \frac{1}{A_{\text{ref,MLA}}} \right)^2$$

where:

- A_{MLA} is the observed minimum lumen area;
- $A_{\text{ref,MLA}}$ is either the lumen-envelope reference area or the Glagov reference area at the MLA location;
- K_i is the adopted contraction/separation coefficient, with primary value 1.2 and sensitivity range $[1.0, 1.5]$;
- $\rho = 1050$ kg/m³.

For lumen-only models, $A_{\text{ref,MLA}} = A_{\text{ref,L}}(s_{\text{MLA}})$. For Glagov models, $A_{\text{ref,MLA}} = A_{\text{ref,G}}(s_{\text{MLA}})$.

This is the physically appropriate location for Glagov information in a reduced-order stenosis model^{18,30}.

In practice, $A_{\text{ref,MLA}}$ is taken from the reconstructed local reference profile at the MLA position rather than from the downstream expansion geometry assumed in the classical Young-Tsai formulation. In tapering vessels or lesions with distal recontraction, this is therefore a reduced-order approximation to the classical Young-Tsai setting rather than an exact geometric recovery.

8.4 Total pressure drop and CT-FFR

$$\Delta P = \Delta P_{\text{visc}} + \Delta P_{\text{sep}}$$

$$\text{CT-FFR} = 1 - \frac{\Delta P}{P_a}$$

where P_a is aortic pressure and Q denotes the hyperemic per-vessel flow $Q_{\text{hyp},v}$ implied by the selected boundary-condition model. For development-phase analyses we use nominal $P_a = 100$

mmHg (13,332 Pa) as a population-independent normative planning constant — the same boundary-condition convention used in the HeartFlow-style reduced-order literature — with sensitivity analysis over 90–110 mmHg spanning the measured mean hyperemic aortic pressures in the FAME, NXT, and DISCOVER-FLOW populations (typically 93–97 mmHg). Resting flow enters only as an intermediate quantity, converted to hyperemic flow via $Q_{\text{hyp},v} = H Q_{\text{rest},v}$.

For development-phase analyses we treat this as a per-vessel reduced-order approximation suitable for direct comparison against invasive FFR; Appendix A2 and Appendix A3 derive the viscous and contraction terms explicitly, and Appendix A4 gives the CT-FFR identity.

9 Flow Allocation Models

9.1 Diameter-based comparator

The diameter-based comparator is included as a pragmatic baseline, consistent with prior reduced-order CT-FFR boundary-condition practice, but serves here as a benchmark rather than as the preferred physiological formulation^{7,16}. Appendix A7 summarizes its Murray-law motivation and the role of the empirical coronary exponent:

$$Q_{\text{rest},v} = Q_0 \left(\frac{D_{\text{prox},v}}{3.0 \text{ mm}} \right)^{2.7}$$

$$Q_{\text{hyp},v} = H Q_{\text{rest},v}$$

with benchmark normalization $Q_0 = 120 \text{ mL/min}$ at a 3.0 mm proximal diameter and primary $H = 3.5^{8,31}$. The value of Q_0 is treated here as a comparator calibration, not as a first-principles physiological constant.

9.2 Primary mass-based model

The primary vessel-mass formulation is:

$$M_v = f_v M_{\text{cal}}$$

$$Q_{\text{rest},v} = q_{\text{rest}} M_v$$

$$Q_{\text{hyp},v} = H Q_{\text{rest},v}$$

where:

- M_{cal} is the calibrated CT-derived LV myocardial mass entering the flow model;
- f_v is the vessel territory fraction;
- q_{rest} is resting perfusion per gram, centered at approximately 1.0 mL/min/g, i.e. approximately $1.67 \times 10^{-8} \text{ m}^3/\text{s/g}$ when converted to SI units⁸;
- H is the hyperemic multiplier^{8,31}.

In general, the calibrated mass map can be written as $M_{\text{cal}} = a + bM_{\text{CT}}$. For the current revision cycle, the stage-1 default is a cohort-wide zero-intercept slope-only Deming calibration $M_{\text{cal}} = bM_{\text{CT}}$ unless the paired CT-CMR subset shows a clinically material intercept large enough to justify freeing a .

The per-gram formulation replaces the total-coronary-flow reference used in earlier reduced-order CT-FFR iterations (which normalised to a single total coronary flow of approximately 250 mL/min). For the current revision cycle, the primary allometric sensitivity anchor is fixed at $M_{\text{ref,cal}} = 250 \text{ g}$, so the per-gram and historical total-flow forms agree at that reference point (where $q_{\text{rest}} \approx 1.0 \text{ mL/min/g}$ gives $Q_{\text{total}} \approx 250 \text{ mL/min}$). They then diverge intentionally in patients with abnormal LV mass: the per-gram formulation reduces demand correctly in post-infarction patients with reduced viable mass and increases it appropriately in patients with left-ventricular hypertrophy.

9.2.1 Sex-specific reference LV mass

Because CT-derived LV mass is used as a plausibility check and sensitivity context, sex-specific reference ranges remain useful for interpreting individual patient values and identifying outliers even though the primary allometric anchor is the cohort-wide constant $M_{\text{ref,cal}} = 250 \text{ g}$. CMR-based population reference values from the UK Biobank provide the best-powered normative data currently available³²:

Sex	Reference LV mass (CMR)	Interquartile range	Source
Female	~95 g	~78–113 g	³² (n ≈ 4,400)
Male	~140 g	~116–165 g	³² (n ≈ 4,400)

These reference values serve three narrower roles in the present programme:

1. **Plausibility filter:** CT-derived M_{cal} values outside approximately 50–200 g (female) or 70–250 g (male) are flagged for review before entering the flow model.
2. **Sensitivity context for calibration:** the primary calibration map remains cohort-wide in the current revision cycle, but sex-specific residual patterns may be inspected as a plausibility or sensitivity analysis if calibration drift by sex appears material.
3. **Exploratory subgroup:** sex-stratified mass-flow behavior will be reported descriptively to assess whether mass-based demand allocation behaves consistently across the sex split of the DISCHARGE cohort.

9.3 Territory fractions

Because measured territory maps will not be available for most development-phase cases, the primary model uses anatomical territory fractions with dominance adjustment:

Dominance	LAD	LCx	RCA
Right-dominant	0.42	0.30	0.28
Left-dominant	0.42	0.38	0.20
Co-dominant	0.42	0.33	0.25

This is explicitly an anatomical approximation, not a direct perfusion measurement. The LAD fraction is held approximately constant because dominance is modeled mainly as redistribution of inferior and posterolateral supply between LCx and RCA rather than of anterior LV supply; this is consistent with coronary flow-volume data showing larger dominance-related shifts in LCx and RCA flow, with relatively stable LAD flow across dominance patterns³³. The use of territory mass is further supported by human data showing that CT-derived myocardial mass tracks invasive relative perfusion in proximal coronary territories³⁴.

9.4 Allometric sensitivity analysis

To preserve the original mass-scaling concept while keeping the primary model simple, we define:

$$Q_{\text{rest},v}(\gamma) = q_{\text{rest}} M_{\text{ref},v} \left(\frac{M_v}{M_{\text{ref},v}} \right)^\gamma$$

with

$$M_{\text{ref},v} = f_v M_{\text{ref,cal}}$$

and, for the primary sensitivity branch, $M_{\text{ref,cal}} = 250$ g. The normalization ensures that the allometric model matches the primary linear model at $M_v = M_{\text{ref},v}$ for all γ , and the model reduces to $Q_{\text{rest},v} = q_{\text{rest}} M_v$ when $\gamma = 1$.

This makes the primary linear model the $\gamma = 1.0$ case, which is the simplest human-development approximation when perfusion is expressed in mL/min/g. The classical allometric model is retained as the lower-bound sensitivity case at $\gamma = 0.75$ ^{8,9,35}.

9.5 Hyperemia handling

Primary model:

- fixed $H = 3.5$ for development-phase comparison^{8,31}.

This value is derived from healthy-volunteer resting and hyperemic MBF data ($H \approx 3.54 \pm 1.01$ from PET 15O-water;⁸) and is appropriate as a normative development-phase constant when invasive flow measurements are not available.

Sensitivity analysis:

- $H \in [2.5, 4.0]$, spanning approximately ± 1 SD around $H = 3.5^8$.

9.5.1 Disease-stratified exploratory H values

In the presence of significant microvascular disease, the hyperemic multiplier is attenuated.³⁶ demonstrate that FFR–CFVR discordance reflects predominantly microvascular impairment, and that coronary flow velocity reserve (CFVR) is progressively reduced in proportion to microvascular burden. Drawing on that framework, the following disease-stratified H values are available as exploratory inputs for sensitivity analyses in the prognostic stage:

Disease state	Exploratory H	Basis
No significant CAD / healthy reference	3.5	⁸ (healthy volunteers)
Intermediate CAD with microvascular component	2.8	Inferred from ³⁶ : CFVR reduction in FFR–CFVR discordant group
Severe CAD with significant microvascular impairment	2.5	Modeled assumption (lower bound of primary sensitivity range; no direct reference)

These values are not embedded in the primary model. They are reserved for exploratory analyses in the prognostic stage where disease severity stratification is available, and the $H=2.5$ value must be explicitly labeled as a modeled assumption with no direct reference support. Disease-specific microvascular adjustment requires direct invasive flow measurements to be validated; the present development phase does not include those measurements.

10 Validation Framework

10.1 Common-case paired target-vessel subset

The primary diagnostic analysis is not run on all recorded CT or all recorded FFR cases. It is run on the **strict four-arm complete-case paired target-vessel subset** that survives CTA matching, lesion mapping, stage-1 geometry QC, and has all four model inputs available. The working inclusion/exclusion logic is:

Rule	Stage-1 operational default	Consequence
Matched CTA and invasive FFR for the same target vessel	required	Include in primary paired subset only if vessel matching is credible
Continuous analyzable vessel span	proximal reference span ≥ 5 mm and distal interrogation span ≥ 10 mm from the MLA	Exclude from primary paired subset if unresolved after correction
Interpretable MLA neighborhood	no severe unresolved motion, blooming, or branch-overlap obscuration at the MLA	Exclude from primary paired subset if violated
Lumen continuity and ordering	no unresolved gap > 2 mm and no arc-length ordering failure	Exclude from primary paired subset if violated
Bifurcation and stent interpretability	no unresolved multi-lumen bifurcation MLA and no uninterpretable stent-crossing lesion span	Exclude from primary paired subset if violated
Level-0 wall-supported MLA neighborhood	required for primary four-arm subset	If absent, retain only for sensitivity analyses not requiring primary Glagov-mode inference
Locked acceptable LV-mass state	calibrated or manually_corrected required for primary four-arm subset	If absent, retain only for sensitivity analyses not requiring primary mass-mode inference
Minimum analyzable caliber	equivalent diameter < 1.5 mm at the MLA neighborhood, or persistently < 1.5 mm across most of the distal span	Mark as anatomically too small / hypoplastic and exclude from stage-1 flow analysis

These rules define the strict four-arm complete-case primary subset. A broader matched-vessel pool may still contribute to pre-specified pairwise sensitivity analyses, but it is not the primary inferential set.

The machine-readable QC defaults behind this shorthand are inherited from `segment_v1.md`, espe-

cially the sections Coronary-Specific Strategy, Flow-ready export contract, Stage-1 default QC thresholds, and Analyzability decision logic. If the concise manuscript summary and the segmentation reference ever differ in detail, the segmentation reference defines the authoritative stage-1 export logic for the current revision cycle.

10.2 Primary endpoint

Primary endpoint: per-vessel diagnostic agreement with invasive FFR in the FFR-linked development cohort.

Primary metrics:

- AUC for FFR ≤ 0.80 ;
- mean absolute error versus invasive FFR;
- Bland-Altman bias and limits of agreement;
- calibration slope and intercept;
- threshold agreement at 0.80.

Given CT-to-FFR matching losses, lesion-site ambiguity, and strict geometry QC, we conservatively anticipate approximately 70–90 analyzable paired target-vessel comparisons in the broader matched-vessel pool. For planning purposes, the strict four-arm complete-case primary subset is expected to retain roughly 45–60 paired vessels once direct Level-0 MLA wall support and an acceptable locked LV-mass state are both required. This is sufficient for development-phase model ranking, interval estimation, and effect-size reporting, but not for claiming confirmatory power for very small differences between neighboring reduced-order variants.

10.3 Power and detectable-effect analysis

10.3.1 Diagnostic arm (FFR-linked cohort)

Planning assumptions

Parameter	Value	Basis
Broad matched-vessel planning pool	$n = 80$	Conservative: 113 recorded cases minus CT-to-FFR matching losses, site ambiguity, and geometry QC failures
Ischemic prevalence (invasive FFR ≤ 0.80)	35–40%	Stable CAD referral pattern in DISCHARGE; planning value 37.5% $\rightarrow n_+ \approx 30, n_- \approx 50$

Parameter	Value	Basis
Baseline AUC (L-CT-FFR vs invasive FFR)	0.75	Conservative; published reduced-order CT-FFR ROC AUC range 0.72–0.82
Inter-model correlation	$r = 0.75\text{--}0.80$	Models share anatomy; partial correction only
Significance level	nominal $\alpha = 0.05$, one-sided	Planning approximation before Holm correction of the three primary AUC contrasts
Target power	80%	

AUC planning approximation (Hanley-McNeil variance with paired-correlation adjustment)

Using the Hanley-McNeil large-sample AUC variance approximation³⁷ with $n_+ = 30$, $n_- = 50$, $AUC_0 = 0.75$:

$$\text{Var}(\widehat{AUC}) = \frac{AUC(1 - AUC) + (n_+ - 1)(q_1 - AUC^2) + (n_- - 1)(q_2 - AUC^2)}{n_+ \cdot n_-} \approx 0.00348$$

where $q_1 = AUC/(2 - AUC) = 0.60$ and $q_2 = 2 AUC^2/(1 + AUC) = 0.643$ are the Hanley-McNeil auxiliary quantities (using lower-case q to avoid clash with hemodynamic flow Q).

For the paired planning contrast, with inter-method correlation $r = 0.75\text{--}0.80$:

$$SE(\Delta\widehat{AUC}) = \sqrt{2 \text{Var}(\widehat{AUC}) (1 - r)} \approx 0.037\text{--}0.042$$

This planning step uses Hanley-McNeil only for a large-sample variance benchmark and then imposes a plausible paired-correlation range r as a design surrogate; it is not the final paired-comparison test. The locked primary inferential procedure for observed model contrasts remains the paired DeLong framework³⁸.

At 80% power, nominal one-sided $\alpha = 0.05$ (critical multiplier $z_{0.05} + z_{0.20} = 2.49$):

$$\Delta AUC_{\min} = 2.49 \times SE(\Delta\widehat{AUC}) \approx 0.09\text{--}0.10$$

Summary: in the broad matched-vessel planning pool, the diagnostic arm has 80% power to detect nominal $\Delta AUC \approx 0.09\text{--}0.10$ between paired models under this planning approximation.

This nominal benchmark is not the final strict-primary decision threshold: the locked primary family applies Holm adjustment across three AUC contrasts on the smaller four-arm complete-case subset, so the true confirmatory threshold will be larger. Under Holm correction, the first adjusted comparison is tested at one-sided $\alpha_1 = 0.0167$, which raises the nominal detectable paired ΔAUC to approximately 0.11–0.12 even before accounting for any further reduction in the strict complete-case subset. Differences smaller than the unadjusted nominal range are therefore estimation signals only, not confirmatory evidence. They become informative when directionally consistent across the four power-anchored comparison metrics (AUC, MAE, threshold agreement, and net reclassification); Bland-Altman and calibration slope are reported as characterization outputs rather than power-dimensioned contrasts.

Mean absolute error (paired Wilcoxon / t-test)

With per-vessel paired error differences having $\text{SD} \approx 0.07$ and $n = 80$:

$$\Delta\text{MAE}_{\min} = 2.49 \times \frac{0.07}{\sqrt{80}} \approx 0.019$$

The study can reliably resolve paired mean absolute error reductions of approximately **0.02 FFR units** at 80% power.

Threshold classification (binary FFR ≤ 0.80)

With $n_+ \approx 30$ and $n_- \approx 50$ and assuming sensitivity $\approx$ specificity ≈ 0.80 , individual sensitivity and specificity estimates carry 95% confidence intervals of approximately $\pm 14\%$ and $\pm 11\%$ respectively (Wilson intervals³⁹; the widths are assumption-dependent and will widen if true sensitivity or specificity departs from 0.80). Between-model classification change is assessed by McNemar's test⁴⁰; with $n = 80$, the test has 80% power to detect net reclassification when approximately **9–11 vessels** (≈ 11 –14%) change classification direction.

Detectable-effect summary

Metric	Detectable difference at 80% power	Interpretation
AUC	nominal $\Delta\text{AUC} \approx 0.09$ –0.10	Moderate discrimination gain; not marginal variant fine-tuning
Mean absolute error	$\Delta\text{MAE} \approx 0.02$ FFR units	Clinically small but detectable systematic improvement
Sensitivity / specificity	$\pm 14\%$ / $\pm 11\%$ (95% CI width)	Threshold metrics informative only for large shifts
Net reclassification	$\geq 10\%$ of vessels	McNemar detectable at 80% power

This is the correct inferential posture for a staged reduced-order development study. The manuscript is powered for model ranking, interval estimation, and falsification of clearly non-useful variants. It is **not** powered for confirmatory claims about marginal gains between neighboring variants of the same reduced-order family, especially once the strict complete-case primary set and Holm-adjusted three-contrast family are enforced.

10.4 Primary comparison order

The locked primary comparison family is:

1. L-CT-FFR vs G-CT-FFR
2. L-CT-FFR vs M-CT-FFR
3. L-CT-FFR vs MG-CT-FFR

These three paired DeLong contrasts are evaluated on the strict four-arm complete-case target-vessel subset, with Holm correction across the family. Maximal eligible pairwise subsets are retained only as sensitivity analyses. The repeated-measures geometry, flow-allocation, and geometry $\times$ flow model is a pre-specified supportive secondary inferential analysis rather than part of the primary Holm-controlled family.

10.5 Paired-difference reporting

The primary endpoint defines the per-model diagnostic metrics. For between-model contrasts, the analysis should additionally report paired effect sizes and uncertainty rather than only rank ordering. In particular, it should report paired AUC differences with DeLong covariance estimates or equivalent paired resampling³⁸, together with paired summaries of error and classification change where informative. Holm-adjusted inference applies only to the three locked primary AUC contrasts against L-CT-FFR; all other between-model summaries are descriptive or supportive-secondary unless explicitly promoted later. The development-phase question is whether the geometry and demand corrections move the model in the right direction by a clinically meaningful amount, not whether tiny differences can be declared confirmatory.

10.6 Monte Carlo uncertainty quantification

A formal perturbation-based uncertainty quantification (UQ) pipeline is a named innovation of this framework. For each analyzed vessel, a Monte Carlo UQ procedure perturbs the principal model inputs simultaneously within their empirical or assumed distributions and records the resulting CT-FFR distribution at the vessel level:

Input	Perturbation distribution	Rationale
K_v	Uniform $[8\pi, 32]$	Full sensitivity range from exact to reduced-order coefficient
K_i	Uniform $[1.0, 1.5]$	Borda-Carnot ideal to empirical upper bound

Input	Perturbation distribution	Rationale
H	Normal $\mathcal{N}(3.5, 0.5^2)$, truncated to $[2.5, 4.0]$	Centered on healthy-volunteer data ⁸
P_a	Uniform $[90, 110]$ mmHg	Normative aortic pressure range
T_{healthy}	$\pm 15\%$ of vessel-level estimate	Reflects outer-wall segmentation uncertainty
M_{cal}	$\pm 10\%$ of calibrated value	CT mass measurement and calibration uncertainty

For each vessel, $N = 2,000$ draws are sufficient to estimate the 5th–95th percentile CT-FFR interval with acceptable Monte Carlo error. The primary outputs are:

1. **Vessel-level uncertainty interval:** 5th–95th percentile CT-FFR range, used to flag vessels where the ischemia classification is sensitive to parameter uncertainty.
2. **Classification stability fraction:** proportion of Monte Carlo draws that agree with the point-estimate classification at FFR = 0.80.
3. **Sensitivity decomposition:** variance decomposition by input (Sobol-type index or simplified one-at-a-time sensitivity) to identify which parameter drives most of the output uncertainty.

This UQ pipeline is applied to all analyzable vessels in Cohort A. Results are reported as a supplementary table alongside the primary diagnostic comparison. No clinical decision uses the MC interval directly; it is an analytic transparency tool for the development phase.

10.7 Reproducibility and tool comparison

Where repeated Medis reads exist, inter-reader agreement should be quantified for:

- outer-wall area over trusted intervals;
- reconstructed $A_{\text{ref,MLA}}$;
- final CT-FFR values.

The preferred summaries are ICC-type reliability estimates plus Bland-Altman plots and explicit reporting of the fraction of vessels whose clinical classification changes across readers. Where overlapping alternative-tool segmentations exist, the same comparisons should be repeated across tools rather than silently pooling them into one geometry source.

10.8 Fallback-mode reporting

Every analyzed vessel should be labeled by the reference-support hierarchy defined above and operationalized in Appendix A5.5: direct measured wall (Level 0), local reconstruction (Level 1), reference-lumen or taper prior (Level 2), population/anatomical prior (Level 3), or lumen-envelope reversion. Primary G-CT-FFR and MG-CT-FFR analyses should require Level-0 support across the MLA-centered

lesion/reference interval; Levels 1 and 2 belong in sensitivity analyses, and Level 3 is excluded from primary Glagov-mode analyses. At minimum the manuscript should report the Glagov-eligible rate, fallback-level frequencies, and diagnostic performance in clean Level-0 cases versus fallback-supported subsets.

10.9 FFR subset representativeness

Because invasive FFR referral in DISCHARGE was not random, the FFR-linked cohort should not be treated as an unqualified miniature of the full CT cohort. Baseline age, sex, diabetes, hypertension, CT stenosis severity, and calcium burden should be compared between the FFR-linked cohort and the broader CT dataset. If material imbalances persist, weighted or stratified sensitivity analyses should accompany the primary diagnostic results.

10.10 Reproducibility and robustness sensitivity

Secondary technical analyses:

- perturbation of lumen and outer-wall contours;
- inter-reader agreement where repeated Medis segmentations exist;
- Medis vs alternative segmentation-tool comparison where overlapping data are available;
- global parameter sensitivity of CT-FFR to $K_v \in [8\pi, 32]$, $K_i \in [1.0, 1.5]$, $H \in [2.5, 4.0]$, and $P_a \in [90, 110]$ mmHg, with $\gamma \in [0.75, 1.0]$ evaluated for the allometric mass-scaling variant;
- robustness of CT-FFR to fixed vs dominance-adjusted territory fractions;
- clean-Level-0 versus fallback-supported subset analyses for the reference hierarchy.

10.10.1 1D network credibility cross-check

As an independent credibility check on 50–100 cases from the FFR-linked cohort, the G-CT-FFR variant will be re-evaluated using a 1D network solver (SimVascular svOneDSolver;⁴¹) with Glagov reference geometry and identical boundary conditions. The role of this cross-check is verification, not replacement: systematic disagreement > 0.05 in mean CT-FFR between the reduced-order analytic result and the 1D network result triggers case-by-case review to identify geometry or boundary-condition inconsistencies. Agreement confirms that the analytic approximation is an adequate summary of the distributed 1D pressure-drop calculation for the vessel spans included in this study.

10.11 Exploratory prognostic stage

Patient-level prognostic modeling is retained, but reframed:

- exploratory;
- contingent on obtaining a sufficiently harmonized patient-level CT subset;
- not the primary endpoint for method development;
- not used as a decision criterion for the core model development.

If a sufficiently harmonized cohort becomes available, a parsimonious patient-level model may still be reported as hypothesis-generating, ideally with CT-FFR, anatomical stenosis, and a minimal covariate set chosen to respect the limited event count. This preserves the long-term clinical ambition without collapsing the development-phase and prognostic questions into one claim.

The same caution applies to subgroup claims. With a low-event exploratory prognostic dataset, the following subgroup analyses should be treated as descriptive only:

Exploratory subgroup	Expected information level	Interpretation
Left-ventricular hypertrophy	low event burden	severely underpowered for interaction testing
Prior infarction / scar	very low event burden	severely underpowered for interaction testing
Diabetes	low event burden	underpowered for interaction testing
Sex	modest split of a low-event cohort	underpowered for interaction testing
High calcium burden	low event burden	severely underpowered for interaction testing

Subgroup hazard ratios may still be reported with wide confidence intervals, but no subgroup should be used to support a confirmatory claim unless the signal is both large and directionally consistent with the primary analysis.

10.11.1 Exploratory prognostic power

An explicit event-count audit is required here because the DISCHARGE MACE definition is narrow: the primary composite is **cardiac death + non-fatal MI + non-fatal stroke** at 3.5-year median follow-up. The published CT arm ($n \approx 1,808$) recorded approximately **38 MACE events** (rate $\sim 2.1\%$), not the 8–12% rates typical of broader composite endpoints that include revascularization or hospitalization.

Planning assumptions

Parameter	Value	Basis
CT cohort with outcome follow-up	$n \approx 1,800$	DISCHARGE CT arm
MACE definition	Cardiac death + non-fatal MI + non-fatal stroke	DISCHARGE primary composite
Observed MACE rate at 3.5 years	$\sim 2.1\%$	DISCHARGE publication

Parameter	Value	Basis
Expected events	~38	Directly from DISCHARGE CT arm
Max prespecified model parameters (EPV ≤ 3 ≥ 10)		38 / 3 = 12.7; conservative EPV target ^{42,43} fails for ≥ 4 predictors

What 38 events implies

Using Schoenfeld's equal-allocation approximation for the binary CT-FFR contrast (FFR ≤ 0.80 vs > 0.80 ; two-sided $\alpha = 0.05$, 80% power; equal allocation $p = 0.5$)⁴⁴:

$$d = \frac{4(z_{\alpha/2} + z_{\beta})^2}{[\ln(\text{HR})]^2} \Rightarrow \ln(\text{HR})_{\min} = \sqrt{\frac{31.4}{38}} = 0.91 \Rightarrow \text{HR}_{\min} \approx 2.5$$

A binary contrast is therefore only reliably detectable for $\text{HR} \geq 2.5$, which is clinically extreme and unlikely for a CT-FFR threshold in stable CAD. Because this is an equal-allocation approximation, unequal CT-FFR group sizes would require an even larger detectable hazard ratio. The FFR-linked development subset is also an invasive-enriched diagnostic cohort rather than a population sample, so the realized CT-FFR split may deviate materially from $p = 0.5$; this makes the equal-allocation benchmark mildly optimistic and suitable only as an order-of-magnitude screen. For a **continuous** CT-FFR predictor the same event count still implies that only large standardized effects would be detectable; no dedicated power derivation is prespecified here beyond that order-of-magnitude conclusion. **No between-model ΔC -statistic comparison is powered or prespecified**: survival-discrimination contrasts require a separate framework such as Uno-type censored-data C-statistics with resampling⁴⁵, and with only 38 events they are too unstable to serve as a meaningful design target.

Honest capability at 38 events

Analysis	Feasibility	Interpretation
Crude univariable association of CT-FFR with MACE	Feasible; HR and 95% CI will be wide	Directional signal only
Multivariable Cox with ≤ 3 predictors	Borderline (EPV ≈ 12 –13)	Exploratory; results are hypothesis-generating
Binary CT-FFR prognostic stratification (HR ≥ 2.5)	Powered only for large effects	Clinically implausible threshold; not a design target

Analysis	Feasibility	Interpretation
Nested model comparison (MG-CT-FFR vs L-CT-FFR)	Not powered	No prespecified ΔC design target; survival-discrimination contrasts are too unstable at 38 events
Subgroup interaction testing	Severely underpowered for all subgroups	Descriptive reporting only

Conclusion: the prognostic stage as constrained by the DISCHARGE narrow-MACE event count is **not powered for model comparison or confirmatory CT-FFR prognostic claims**. It can contribute a crude, wide-CI univariable association estimate. This reinforces the reframing throughout this section: the prognostic stage is hypothesis-generating only, and the diagnostic (FFR-validation) arm carries the study's primary inferential weight.

Subgroup power is intentionally not computed. All subgroup analyses should be treated as purely descriptive; HR estimates should be reported with 95% CI and interaction p-values, with explicit acknowledgement that the event count precludes any interaction testing.

If sufficient Level-0 MLA support exists, RI_Glagov may also be summarized numerically in that same subset and compared descriptively with the paired difference between Glagov-based and lumen-envelope CT-FFR. No confirmatory threshold claim or category table is locked for the initial manuscript.

11 Implementation Roadmap

11.1 Phase 1: Cohort assembly and geometry QC (Months 1–4)

- match invasive FFR cases to CT and vessel exports;
- define analyzability criteria;
- build the FFR-linked development cohort;
- harmonize lumen and outer-wall area profiles.

11.2 Phase 2: Core model development (Months 5–9)

- implement the 2×2 model family;
- lock the corrected contraction term;
- run initial diagnostic comparison in the FFR-linked cohort;
- refine outer-wall fallback logic and territory-fraction sensitivity.

11.3 Phase 3: Hierarchical reference-reconstruction strategy for unreliable outer-wall data (Months 10–12)

- define trusted wall intervals and MLA-support rules;

- implement Level-1 and Level-2 reference-lumen fallback modes as explicit sensitivity analyses;
- lock the stop rule for Level-3 priors and lumen-envelope reversion;
- quantify the Glagov-eligible rate and fallback-support frequencies.

11.4 Phase 4: Robustness and reproducibility (Months 13–16)

- inter-reader and segmentation-tool sensitivity;
- contour perturbation analyses;
- clean-Level-0 versus fallback-supported subset analyses;
- subgroup analyses by calcium burden and vessel type.

11.5 Phase 5: Expanded processing and exploratory prognosis (Months 17–24)

- scale to the broader harmonized CT subset;
- compute exploratory patient-level prognostic associations;
- prepare manuscripts and external-validation planning.

The extended Phase-5 horizon reflects the additional time required for explicit fallback-hierarchy development and for broader cohort harmonization beyond the FFR-linked development set. Prognostic analyses remain secondary to the core method-development work.

12 Discussion

12.1 Relation to existing CT-FFR approaches

This framework occupies a position among three established CT-FFR families rather than outside them. Relative to full 3D CFD pipelines, it sacrifices some fluid-dynamic fidelity but gains transparency, speed, and feasibility for repeated sensitivity analyses¹⁶. Relative to reduced-order analytic approaches of the kind evaluated in the FAVOR studies, it retains the pressure-drop logic while making reference geometry an explicit modeling target rather than a tacit lumen-derived assumption^{17,18}. Relative to machine-learning surrogates, it keeps the mapping from geometry and flow demand to pressure loss explicit and auditable^{19,20}.

The intended claim is therefore narrow. The manuscript does not argue that current CT-FFR methods are useless, nor that reduced-order physics is intrinsically superior to 3D CFD. It argues that a transparent reduced-order setting is the right place to test two specific corrections that existing lumen-centered pipelines cannot isolate cleanly: remodelled-vessel reference reconstruction and demand allocation from myocardial mass.

12.2 What is already supported and what DISCHARGE still has to test

Component	Current evidence status	What DISCHARGE is designed to test
Positive remodeling can preserve lumen caliber	Well supported by pathology and IVUS literature ^{4–6}	Whether reference reconstruction changes reduced-order CT-FFR meaningfully in the analyzable DISCHARGE subset
Reduced-order pressure-drop coefficients are plausible closures	Supported by Young-Tsai reduced-order stenosis formulation ^{30,18} extend with turbulence terms for intermediate stenoses, establishing the range of applicability	Whether the corrected contraction-term placement and empirical K_i , K_v perform robustly in the DISCHARGE geometry
Myocardial mass is a more physiological driver of demand than vessel caliber alone	Supported by myocardial mass-flow scaling and perfusion-per-gram literature ^{8,9}	Whether vessel-territory mass allocation improves diagnostic ranking over diameter-based allocation
Kernel-smoothed or prior-based reference reconstruction can approximate a healthy lumen profile	Engineering precedent exists ^{27,28}	Whether such reconstructions are useful only as sensitivity analyses or can support a broader analyzability set without overclaiming
Fallback hierarchies can preserve workflow continuity	Methodologically plausible	Whether the hierarchy improves robustness without diluting the interpretation of primary Glagov-mode results

This distinction matters because the manuscript does not ask DISCHARGE to prove every mechanistic premise from zero. It asks the study to test which parts of the already plausible framework hold up against the actual vessel data and which should be treated as supplementary.

12.3 The flow-demand paradox and why mass-based allocation helps

Diameter-based CT-FFR faces a structural paradox in remodelled vessels. If one acknowledges that the healthy reference vessel was larger than the observed lumen, should flow demand also be in-

creased? Using the observed diameter risks underestimating demand in remodelled arteries, whereas using the reconstructed reference diameter as a demand surrogate risks overstating flow in a territory whose myocardial mass has not changed. Myocardial mass breaks that circularity because it lets resistance come from vessel geometry while demand comes from tissue burden rather than from vessel caliber itself. Although LV mass can itself reflect coexisting hypertrophy or prior injury, it is measured from the CT independently of the target-vessel reference reconstruction and therefore does not inherit the same geometry-demand circularity; any residual confounding from hypertrophic remodeling is deferred to subgroup analyses.

Using a schematic reference territory mass of 150 g, the practical distinction can be illustrated numerically:

Clinical scenario	Example territory mass	$M_v / M_{\text{ref},v}$	Primary demand multiplier ($\gamma = 1.0$)	Allometric sensitivity ($\gamma = 0.75$)	Interpretation
Positive remodeling without mass change	150 g	1.00	$\times 1.00$	$\times 1.00$	Reference geometry may increase while demand stays unchanged
Left-ventricular hypertrophy	225 g	1.50	$\times 1.50$	$\times 1.36$	Higher tissue burden should increase expected flow demand

Clinical scenario	Example territory mass		Primary demand multiplier ($\gamma = 1.0$)	Allometric sensitivity ($\gamma = 0.75$)	Interpretation
		$M_v/M_{\text{ref},v}$			
Prior infarction / scar	120 g	0.80	$\times 0.80$	$\times 0.85$	Lower viable mass should reduce expected flow demand

These values are didactic, not cohort estimates. Their purpose is to show why the mass-based branch changes demand in clinically plausible directions while keeping the primary model ($\gamma = 1.0$) and the allometric sensitivity branch ($\gamma = 0.75$) conceptually separate.

The practical consequence is not that mass-based flow is guaranteed to win in every subgroup. It is that the geometry effect and the demand effect can finally be tested separately within the 2×2 design rather than being conflated inside one lumen-derived boundary condition.

12.4 Why remodeled and bifurcation lesions are strategically important

The reference problem is most acute in lesions where lumen appearance is a poor surrogate for pre-disease caliber. Positively remodelled plaques and bifurcation-adjacent disease are therefore not niche edge cases but strategically informative stress tests for the framework. They are also the settings in which explicit bifurcation-adjacent exclusion zones, local fallback logic, and MLA-centered trust intervals are most important. A method that cannot state how it behaves there will remain difficult to interpret even if its aggregate AUC is acceptable.

12.5 Virtual pullback and spatially resolved pressure profiles

The distributed 1D reduced-order model yields more than a single distal CT-FFR value: it computes a full pressure profile $P(s)$ along the vessel arc from proximal reference to distal interrogation point. This spatially resolved output enables virtual pullback analysis as a secondary analytic output, without additional computation.

Clinical motivation. Invasive physiological pullback traces are increasingly used to phenotype lesions as focal versus diffuse before revascularization decisions⁴⁶. Focal disease (high pullback pressure gradient, PPG) is associated with specific plaque phenotypes and responds well to PCI with high post-intervention FFR gain; diffuse disease (low PPG) predicts suboptimal physiological recovery from stenting and often reflects significant microvascular co-disease^{47,48}. A CT-based virtual pullback profile could extend this phenotyping to the noninvasive setting.

Model output. From the distributed viscous integral, the cumulative pressure at any arc-length position s_k is:

$$P(s_k) = P_a - K_v \mu Q \int_0^{s_k} \frac{ds}{A_{lum}(s)^2} - \Delta P_{sep}(s_k),$$

with

$$\Delta P_{sep}(s_k) = \Delta P_{sep} \mathbf{1}[s_k \geq s_{MLA}],$$

so the contraction loss enters as a step decrease at the MLA position. The point-to-point pressure gradient $dP/ds \propto A_{lum}(s)^{-2}$ directly maps the spatial distribution of viscous resistance along the vessel, with high gradient at the stenosis throat and low gradient in reference segments. A PPG-style summary index as used in the invasive pullback literature⁴⁶ is computable from this profile as a secondary output.

Development-phase status. Virtual pullback profiles will be computed for all analyzed vessels in Cohort A as an exploratory output. They will not be used to modify the primary diagnostic comparison; they are analytic transparency tools that may support future lesion-specific analyses and provide a basis for direct comparison with invasive pullback data where available in the DISCHARGE FFR subset. Formal validation of CT-derived PPG against invasive pullback is deferred to a dedicated future study.

12.6 Why the fallback hierarchy strengthens rather than weakens the study

Explicit fallback modes are a strength because they force the project to separate evidence levels rather than hide them. A vessel supported by directly observed outer-wall data is not the same as one supported by kernel-smoothed reference reconstruction or by a taper prior, and the manuscript reports that difference cleanly. This reduces dependence on universal outer-wall success without pretending that every fallback mode carries the same physiological credibility.

13 External Validation Plan

Internal DISCHARGE validation establishes the development-phase diagnostic benchmark. External validation against an independent cohort is the pre-specified next step before any clinical translation claim.

Target cohorts for external validation:

Cohort	Trial size / CT-usable pool	FFR subset	Outer-wall data	Myocardial mass	Status
SCOT- HEART ⁴⁹	~4,100 randomised total; CT-derived analyses would use the CTA-imaged arm	Invasive FFR in subset	Not systematically available	Not systematically available	Pre-specified target; CT available
PROMISE ⁵⁰	~4,900 CT arm	Invasive FFR in subset	Not systematically available	Possible from CT volumetrics	Pre-specified target

Validation scope. External validation will first apply the pre-specified best-performing model from the internal DISCHARGE development phase to the external cohort's available CT and FFR data. Because outer-wall and mass data may not be uniformly available in external cohorts, the lumen-only model (L-CT-FFR) is the primary transferable benchmark; Glagov-mode and mass-mode performance will be reported where data permit.

Timing. External validation is not part of the current grant scope; it is the logical next phase following successful internal diagnostic validation and code publication. A formal data access request to the SCOT-HEART study group will be initiated after the primary DISCHARGE results are available.

14 Limitations

1. Partial vessel segmentations remain an active work package; development-phase CT-FFR is therefore initially limited to analyzable target-vessel spans rather than complete coronary trees.
2. Territory fractions are anatomically assigned for most cases and therefore approximate.
3. Outer-wall quality will be lowest in calcified segments and will require local reversion to lumen-envelope mode where delineation is unreliable.
4. The contraction term is centered on a single dominant MLA; serial or tandem lesions within one vessel span are not yet handled explicitly; vessels with tandem disease should be deferred to a dedicated multi-lesion extension²⁷.
5. The contraction term is still reduced-order; 3D flow separation is represented only approximately through K_i .
6. Prognostic analysis in DISCHARGE should be treated as exploratory until a sufficiently large harmonized cohort is available.
7. External validation against SCOT-HEART or PROMISE is pre-specified but outside the current grant scope.

15 Conclusion

This manuscript describes a staged reduced-order CT-FFR development program with invasive FFR validation as the primary endpoint, partial vessel segmentation treated as an active work package, and a model family that separates geometry effects from demand effects.

The key mathematical correction is explicit:

- actual lumen geometry enters the viscous term;
- reference geometry enters the contraction term;
- mass-based flow is allocated at the vessel-territory level, not by assigning total LV demand to a single vessel.

This framework is technically cleaner and more transparent in how geometric and demand corrections enter the pressure-drop model. Whether it is physiologically more defensible than existing approaches is precisely the empirical question the study is designed to test by comparison against invasive FFR. If validated, correcting both the reference geometry and the demand boundary condition has the potential to reclassify a clinically meaningful fraction of borderline lesions — most plausibly in patients with positive remodelling, left-ventricular hypertrophy, or prior infarction, where both assumptions are most likely to be simultaneously violated. That potential is the scientific motivation; the study is designed to test it, not to assert it.

16 Appendix A: Derivation and Model Closure

This appendix separates four different mathematical statuses that should not be conflated:

1. **Exact derivation**: obtained directly from the stated governing equations.
2. **Reduced-order closure**: a simplified constitutive or loss relation introduced to make the model usable.
3. **Empirical prior or calibration**: a numerical value anchored to prior literature or internal calibration rather than algebra alone.
4. **Algorithmic definition**: a practical estimator or processing rule rather than a physical law.

This distinction is necessary if the model claims are to remain mathematically defensible.

16.1 A1. Notation and units

The core quantities are:

- s : vessel arc length [m];
- L : analyzed vessel length [m];
- $A(s)$: lumen cross-sectional area [m²];
- $r(s)$: lumen radius [m];
- Q : volumetric flow [m³/s], interpreted in the CT-FFR equations as hyperemic per-vessel flow $Q_{\text{hyp},v}$;

- μ : dynamic viscosity [Pa s];
- ρ : blood density [kg/m^3];
- P_a : aortic pressure [Pa];
- P_d : distal pressure [Pa];
- M_{cal} : calibrated left-ventricular myocardial mass entering the flow model [g];
- M_v : vessel territory mass [g];
- f_v : vessel territory fraction [dimensionless];
- q_{rest} : resting perfusion per gram [mL/min/g], with $1.0 \text{ mL/min/g} = 1.67 \times 10^{-8} \text{ m}^3/\text{s/g}$ in SI units;
- H : hyperemic multiplier [dimensionless].

Dimensional consistency checks used below are straightforward:

- $\mu Q \int ds / A^2$ has units Pa;
- $\rho Q^2 / A^2$ has units Pa;
- $q_{\text{rest}} M_v$ has units mL/min.

16.2 A2. Distributed viscous pressure drop

16.2.1 A2.1 Exact circular-tube derivation

For steady, incompressible, fully developed laminar flow in a circular tube of local radius $r(s)$, Hagen-Poiseuille theory gives the differential pressure gradient

$$\frac{dP}{ds} = \frac{8\mu Q}{\pi r(s)^4}.$$

Integrating along the analyzed centerline yields

$$\Delta P_{\text{visc}}^{\text{circ}} = \frac{8\mu Q}{\pi} \int_0^L \frac{ds}{r(s)^4}.$$

Because $A(s) = \pi r(s)^2$, we have

$$r(s)^4 = \frac{A(s)^2}{\pi^2}.$$

Substituting gives the exact circular area-form identity

$$\Delta P_{\text{visc}}^{\text{circ}} = 8\pi\mu Q \int_0^L \frac{ds}{A(s)^2}.$$

Therefore the exact first-principles area-form coefficient is

$$K_v^{\text{circ}} = 8\pi \approx 25.13,$$

not 32.

16.2.2 A2.2 Manuscript form and closure interpretation

The manuscript uses

$$\Delta P_{\text{visc}} = K_v \mu Q \int_0^L \frac{ds}{A_{\text{lum}}(s)^2}$$

with $K_v = 32^{30}$. The exact derivation above justifies the functional dependence $\mu Q \int ds/A^2$, but not the numerical identity $K_v = 32$.

The correct interpretation is that $K_v = 32$ is an adopted reduced-order closure coefficient. It absorbs the departure of real coronary lesions from the ideal assumptions behind Poiseuille flow, including:

- non-circular cross-sections;
- profile distortion near stenoses;
- entrance and exit effects;
- local curvature and asymmetric geometry;
- the practical convention used in the Young-Tsai reduced-order stenosis literature.

So the viscous term has an exact structural derivation and an empirical closure coefficient.

Note on¹⁸: Lyras and Lee (2021) propose an *improved* reduced-order model that adds an explicit turbulence dissipation term precisely because the standard two-term viscous + contraction form underestimates pressure drop in intermediate and severe stenoses¹⁸. Their paper therefore characterizes the *range of applicability* of the two-term form rather than endorsing $K_v = 32$ specifically. In the present work, the two-term form is used as a transparent, auditable reduced-order development baseline in the moderate-stenosis range; the Lyras turbulence term is noted as a named extension if validation reveals systematic underestimation at more severe lesions.

16.3 A3. Reference-dependent contraction and separation loss

Let the local reference velocity and throat velocity be

$$u_{\text{ref}} = \frac{Q}{A_{\text{ref,MLA}}}, \quad u_{\text{MLA}} = \frac{Q}{A_{\text{MLA}}}.$$

The velocity jump across the contraction is therefore

$$\Delta u = u_{\text{MLA}} - u_{\text{ref}} = Q \left(\frac{1}{A_{\text{MLA}}} - \frac{1}{A_{\text{ref,MLA}}} \right).$$

In an ideal Bernoulli acceleration, kinetic-energy change is reversible. The clinically relevant extra loss across a stenosis is the irreversible part associated with jet formation and incomplete downstream pressure recovery. In reduced-order form, that loss is modeled as proportional to the kinetic-energy density of the velocity jump:

$$\Delta P_{\text{sep}} = K_i \frac{\rho}{2} (\Delta u)^2.$$

Substituting the expression for Δu gives

$$\Delta P_{\text{sep}} = K_i \frac{\rho}{2} Q^2 \left(\frac{1}{A_{\text{MLA}}} - \frac{1}{A_{\text{ref,MLA}}} \right)^2.$$

This explains why Glagov information enters the contraction term rather than the viscous term: the contraction loss depends on the local difference between observed throat area and reconstructed reference area. As with K_v , the structure is mechanistically justified, but K_i remains an empirical reduced-order closure. In the present manuscript, primary analyses use $K_i = 1.2$ with sensitivity range $[1.0, 1.5]$ to represent an order-unity contraction-loss coefficient rather than a first-principles constant³⁰. Note:¹⁸ extend the two-term framework by adding explicit turbulence terms precisely because K_i and K_v as fixed empirical constants are insufficient for intermediate-to-severe stenoses; this extension is noted in A2 and deferred to post-validation development.

The Borda-Carnot ideal corresponds to $K_i = 1.0$, in which the kinetic-energy loss equals exactly the velocity-jump kinetic energy with perfect energy accounting. The adopted primary value $K_i = 1.2$ is an empirical enhancement above that ideal, reflecting that real separated-flow losses typically exceed the Borda-Carnot estimate due to turbulence and three-dimensional jet re-attachment effects. The sensitivity range $[1.0, 1.5]$ spans from the ideal lower bound to an empirical upper bound observed in stenosis-model benchmarks.

Note on prior formulation: earlier reduced-order CT-FFR formulations, including the approach superseded by this framework, expressed contraction loss in a lumped form $K_i(\rho/2)(Q/A_{\text{MLA}})^2$, which implicitly assumes $A_{\text{ref,MLA}} \rightarrow \infty$ (i.e., the reference vessel is infinitely wide upstream). The reference-dependent form here is a deliberate conceptual correction: contraction loss is physically defined relative to a healthy reference vessel, not relative to an infinite reservoir. The lumped form is recovered in the limit $A_{\text{ref,MLA}} \rightarrow \infty$.

16.4 A4. Total pressure drop and CT-FFR

Under the quasi-steady one-dimensional decomposition adopted here,

$$\Delta P = \Delta P_{\text{visc}} + \Delta P_{\text{sep}}.$$

By definition,

$$\Delta P = P_a - P_d.$$

Therefore

$$P_d = P_a - \Delta P.$$

Fractional flow reserve is

$$\text{FFR} = \frac{P_d}{P_a}.$$

Substituting for P_d gives

$$\text{CT-FFR} = \frac{P_a - \Delta P}{P_a} = 1 - \frac{\Delta P}{P_a}.$$

This identity is exact once the reduced-order expression for ΔP has been specified.

For the development-phase manuscript analyses, the pressure-drop model is evaluated at hyperemic flow $Q = Q_{\text{hyp},v}$ and nominal aortic pressure $P_a = 100 \text{ mmHg}$ (13,332 Pa), with sensitivity analysis over 90–110 mmHg.

16.5 A5. Healthy-wall estimator and Glagov reference geometry

16.5.1 A5.1 Primary healthy-wall estimator

The manuscript defines

$$T_{\text{healthy}} = \text{median}(r_{\text{outer}} - r_{\text{lum}})$$

over candidate reference segments. This is not a first-principles physical law. It is an algorithmic estimator for a latent healthy wall thickness. The median is used because it is robust to outliers from noisy contours, local plaque, and calcific blooming.

16.5.2 A5.1.1 Operational candidate-reference criteria

For stage-1 work, the estimator above still needs an explicit rule for which slices are allowed to contribute to T_{healthy} . A practical default is to admit only candidate slices that satisfy all of the following:

- they lie outside the bifurcation-exclusion neighborhood and outside the immediate MLA micro-neighborhood;
- the relative lumen-area gradient remains small,

$$\left| \frac{1}{A_{\text{lum}}} \frac{dA_{\text{lum}}}{ds} \right| \leq 0.05 \text{ mm}^{-1},$$

i.e. no more than about 5% area change per mm;

- outer-wall nesting remains physically ordered on the candidate span, with positive margin on all included slices and interval median margin at least 0.30 mm;
- no severe blooming or motion flag is present, and no outer-wall contour gap larger than about 3 mm traverses the candidate span;
- at least one contiguous proximal or distal support span of length $\geq \max(5 \text{ mm}, D_{\text{loc}})$ — i.e. 5 mm or one local vessel diameter, whichever is larger — remains after those exclusions.

These thresholds are operational QC defaults rather than biological constants. If image-intensity or plaque labels are available, they may be used as additional exclusion flags, but the core candidate-reference rule should not depend on auxiliary information that is absent from a contour-only export. Candidate slices that fail any criterion are excluded. If no qualifying contiguous support span remains after these exclusions, the fallback wall-thickness rule applies.

16.5.3 A5.2 Geometric reference reconstruction

For a locally circular vessel,

$$r_{\text{outer}} = r_{\text{lum}} + T.$$

If the healthy wall thickness at that location is approximated by T_{healthy} , then the lumen radius that would be present without local plaque burden is estimated by subtracting that healthy thickness from the observed outer radius:

$$r_{\text{ref,G}}(s) = r_{\text{outer}}(s) - T_{\text{healthy}}.$$

In the manuscript, T_{healthy} is treated as a vessel-level scalar applied along the analyzed span rather than as a separate profile $T(s)$. This is a direct geometric reconstruction, not an additional hemodynamic assumption.

16.5.4 A5.3 Reference floor safeguard

The manuscript enforces

$$r_{\text{ref,G}}(s) \geq 1.05 r_{\text{lum}}(s).$$

This inequality is not derived from vascular mechanics. It is a geometric and numerical safeguard. Its

role is to prevent unphysical inversions in noisy segments, where a reconstructed “reference” radius could otherwise fall below the observed lumen radius. The 5% radius margin implies an area floor of exactly $1.05^2 = 1.1025$, so the area floor is larger than the underlying radius margin by construction and guarantees a strict ordering between reference and lumen geometry while stabilizing subsequent area-based pressure-drop calculations.

16.5.5 A5.4 Lumen-only comparator

The lumen-only reference

$$r_{\text{ref,L}}(s) = \text{monotone envelope of } r_{\text{lum}}(s)$$

is also an algorithmic definition rather than a derivation. Its purpose is to provide a comparator that uses no outer-wall information while still enforcing physiologic taper.

16.5.6 A5.5 Hierarchical reference-reconstruction fallback

The manuscript distinguishes fallback levels by the support available for the reference geometry:

- **Level 0:** measured outer wall over a trusted MLA-centered interval;
- **Level 1:** local reconstruction from trusted proximal and distal anchors;
- **Level 2:** physiology-constrained or lumen-derived reference-lumen fallback;
- **Level 3:** population or anatomical priors only.

For Level 1, the intended operation is short-gap completion between trusted proximal and distal anchors, for example with shape-preserving interpolation or spline interpolation. It is a local extension of measured support rather than a lumen-only inference.

For Level 2, a generic lumen-derived reference-lumen fallback can be written as a kernel-smoothed estimator on the lumen profile,

$$\hat{r}_{\text{ref}}(s_i) = \frac{\sum_j K_h(s_i, s_j) w_j r_{\text{lum}}(s_j)}{\sum_j K_h(s_i, s_j) w_j},$$

where K_h is a smoothing kernel over arc length and w_j down-weights suspected stenotic points or low-confidence locations. Blanco et al. document a weighted Gaussian-kernel healthy-radius estimator in the supplementary material of their 1D-versus-3D FFR study, itself connected to earlier CTA stenosis-detection work by Shahzad et al.^{27,28}. In the present manuscript this kind of estimator is not treated as recovered outer-wall anatomy, only as a sensitivity-grade reference-lumen fallback.

Other Level-2 options include monotone taper priors or Murray-style branch-order scaling when local anchors are inadequate. Level-3 priors are the weakest support class. If the MLA neighborhood is supported only at Level 3, the vessel is excluded from primary G-CT-FFR and MG-CT-FFR analyses and either reverted to the lumen-envelope comparator or retained only for sensitivity analyses. None

of these fallback levels alters the observed lumen profile in the viscous term; they only affect the support available for the reference-dependent contraction term.

16.5.7 A5.5.1 Artifact-triggered local fallback

The hierarchy above becomes operational only if common failure modes are linked to explicit downgrade rules. A practical stage-1 trigger table is:

Trigger near the MLA interval	Default stage-1 rule	Consequence
Severe blooming or motion on more than 10% of trusted slices	Revoke direct wall support in the affected subspan	Downgrade to Level 1 or Level 2, otherwise revert to lumen-envelope
Outer wall fails physical nesting, e.g. $r_{\text{outer}} \leq r_{\text{lum}} + 0.30 \text{ mm}$ on any trusted slice	Treat wall contour as implausible	Do not use Level 0 support across that interval
Outer-wall contour gap $> 3 \text{ mm}$ crosses the MLA neighborhood	Coverage insufficient for direct Glagov support	Use Level 1 only if bracketing anchors remain; otherwise Level 2 or lumen-envelope
Outer-wall oscillation or jaggedness exceeds the QC limit defined in the segmentation/QC protocol on more than 10% of trusted slices	Geometry judged unstable for direct support	Downgrade the support level
Repeated-reader outer-wall area disagreement $> 20\%$ around the MLA, when repeated reads exist	Treat direct wall evidence as unreliable	Sensitivity-only fallback, not primary Glagov evidence

These are again operational thresholds rather than claims of universal physiological boundaries. Their role is to make the evidence downgrade explicit instead of silently carrying low-confidence wall geometry into $A_{\text{ref,MLA}}$.

16.5.8 A5.5.2 Stage-1 decision logic

The fallback hierarchy can therefore be written as a simple decision sequence:

1. Define the MLA-centered interval after bifurcation-neighborhood exclusion.
2. If measured outer wall covers that MLA-centered lesion/reference interval and passes coverage, nesting, and artifact QC, assign **Level 0** and mark the vessel as primary-eligible for Glagov-mode analysis.
3. Otherwise, if trusted proximal and distal anchors bracket only a short missing span, assign **Level 1** and reconstruct the missing support locally by interpolation.

4. Otherwise, if local anchors are inadequate but vessel identity and physiologic taper constraints remain credible, assign **Level 2** and use a taper prior or kernel-smoothed reference-lumen estimate only as secondary support.
5. Otherwise, revert to the lumen-envelope comparator; reserve **Level 3** population or anatomical priors for sensitivity-only or exclusion-grade analyses.
6. In all non-Level-0 cases, retain the observed lumen profile in the viscous term and do not label the vessel as primary Glagov-supported.

16.5.9 A5.6 Exploratory remodeling index

Classical remodeling indices compare lesion outer-wall area to a nearby reference segment, which becomes ambiguous when adjacent reference tissue is also diseased. For vessels with direct Level-0 wall support at the MLA, the manuscript therefore allows an exploratory model-based remodeling index

$$RI_{\text{Glagov}} = \frac{A_{\text{outer}}(s_{\text{MLA}})}{A_{\text{ref,G}}(s_{\text{MLA}})}.$$

This is an algorithmic derived quantity, not a separate hemodynamic state variable. It is intended to describe remodeling state in the subset where the reference geometry is directly supported by measured wall information.

Operationally, RI_{Glagov} should be computed only when:

- $A_{\text{outer}}(s_{\text{MLA}})$ is available from measured wall data;
- $A_{\text{ref,G}}(s_{\text{MLA}})$ is supported by Level-0 Glagov reconstruction at the MLA neighborhood;
- the vessel is marked `glagov_primary_eligible = true`.

If the MLA neighborhood depends on Level-1 to Level-3 fallback or on lumen-envelope reversion, the correct export state is **indeterminate**, not a forced numerical value.

For descriptive summaries, no remodeling category table is locked in the current manuscript. The denominator is a model-derived reference lumen rather than an independently observed expected outer-wall reference, so RI_{Glagov} is retained as a numeric exploratory summary only.

16.6 A6. Fallback wall-thickness scaling

For a thin-walled cylindrical vessel, Laplace's law gives the approximate circumferential wall stress

$$\sigma_{\theta} \approx \frac{Pr}{T},$$

where P is transmural pressure, r is lumen radius, and T is wall thickness. Rearranging,

$$\frac{T}{r} \approx \frac{P}{\sigma_\theta}.$$

As an approximating assumption rather than a derived necessity, if healthy epicardial vessels operate over a limited pressure range and maintain broadly similar wall stress, then T/r is approximately constant, so

$$T \propto r.$$

Using a representative lumen radius $\tilde{r}_{\text{lum}}$, the fallback rule becomes

$$T_{\text{healthy}} = \beta \tilde{r}_{\text{lum}}.$$

This proportional form is mechanically motivated. However, the numerical choice $\beta \approx 0.18$ is not derived from Laplace's law alone; it is an empirical prior anchored to prior coronary wall-thickness evidence and earlier project assumptions^{24–26}. The additional clip to a total-wall range of 0.40–0.80 mm is likewise a prior chosen to keep CT-visible wall estimates physiologically plausible, with IVUS intima-media measurements interpreted only as physiological anchors rather than as a direct equality with the CTA-visible total-wall contour.

Numerical plausibility check: At a representative proximal coronary with $r \approx 2.5$ mm, $\beta = 0.18$ gives $T_{\text{healthy}} = 0.18 \times 2.5 = 0.45$ mm, which lies inside the applied clip range. Under nominal aortic pressure $P \approx 100$ mmHg = 13,332 Pa, the Laplace circumferential wall stress is

$$\sigma_\theta = \frac{P r}{T} = \frac{13,332 \times 2.5 \times 10^{-3}}{0.45 \times 10^{-3}} \approx 74,000 \text{ Pa} \approx 74 \text{ kPa}.$$

This is consistent with published values for coronary wall stress under physiological loading ($\sigma_\theta \approx 50$ –100 kPa, depending on segment and wall model), confirming that $\beta = 0.18$ is a physiologically plausible Laplace-consistent prior rather than an arbitrary choice.

16.7 A7. Diameter-based benchmark flow model

16.7.1 A7.1 Murray-law motivation

In Murray's classical argument, vascular design balances metabolic maintenance cost of vessel volume against hydraulic pumping cost²⁹. If a vessel segment of length L has radius r , a stylized total cost can be written as

$$C(r) = ar^2L + b\mu Q^2Lr^{-4},$$

where the first term represents maintenance cost and the second viscous pumping cost. Setting $dC/dr = 0$ gives

$$2arL - 4b\mu Q^2 L r^{-5} = 0,$$

so

$$r^6 \propto Q^2.$$

Therefore

$$Q \propto r^3 \propto D^3.$$

This is the theoretical Murray exponent.

16.7.2 A7.2 Manuscript benchmark form

The manuscript benchmark is

$$Q_{\text{rest},v} = Q_0 \left(\frac{D_{\text{prox},v}}{3.0 \text{ mm}} \right)^{2.7}.$$

The exponent 2.7 is therefore an empirical coronary adaptation of the Murray-law ideal 3.0, not an exact first-principles result. The normalization $Q_0 = 120 \text{ mL/min}$ simply fixes the comparator to a chosen proximal diameter reference. Hyperemic flow is then defined by

$$Q_{\text{hyp},v} = H Q_{\text{rest},v},$$

with H treated as an empirical multiplier rather than a derived mechanistic constant.

16.8 A8. Territory-mass flow model

Assume the calibrated left-ventricular myocardial mass is partitioned across vessel territories with fractions f_v satisfying

$$\sum_v f_v = 1.$$

Then

$$M_v = f_v M_{\text{cal}},$$

and summing over vessels gives

$$\sum_v M_v = M_{\text{cal}}.$$

If resting perfusion per gram is q_{rest} , then each territory receives resting flow

$$Q_{\text{rest},v} = q_{\text{rest}} M_v.$$

Summing again,

$$\sum_v Q_{\text{rest},v} = q_{\text{rest}} \sum_v M_v = q_{\text{rest}} M_{\text{cal}}.$$

So the primary mass-based model is exactly the linear mass-partition model implied by:

- conservation of total LV mass across vascular territories;
- use of a common resting perfusion per gram;
- linear conversion from mass to resting flow.

The dominance-adjusted territory fractions f_v that operationalize the partition $\sum_v f_v = 1$ are tabulated by coronary dominance in Appendix A10.

Hyperemic flow is then defined by

$$Q_{\text{hyp},v} = H Q_{\text{rest},v}.$$

Again, the multiplier H is a model choice, not an algebraic derivation.

16.9 A9. Allometric sensitivity model

Suppose resting flow follows a general allometric law

$$Q_{\text{rest},v} = C M_v^\gamma.$$

To make this comparable with the primary linear model, choose a reference territory mass $M_{\text{ref},v}$ at which the allometric model matches the linear model:

$$Q_{\text{rest},v} = q_{\text{rest}} M_{\text{ref},v} \quad \text{when } M_v = M_{\text{ref},v}.$$

That condition implies

$$C = q_{\text{rest}} M_{\text{ref},v}^{1-\gamma}.$$

Substituting back yields

$$Q_{\text{rest},v}(\gamma) = q_{\text{rest}} M_{\text{ref},v} \left(\frac{M_v}{M_{\text{ref},v}} \right)^\gamma.$$

With

$$M_{\text{ref},v} = f_v M_{\text{ref,cal}},$$

the model is anchored to the same territorial decomposition as the primary linear case. Two limits are immediate:

1. $\gamma = 1.0$: the **linear null model**, in which perfusion per gram is constant at q_{rest} across all territory masses. This is the primary model and the most conservative assumption in the absence of validated human-specific territorial scaling data. It recovers $Q_{\text{rest},v}(1) = q_{\text{rest}} M_v$ exactly.
2. $\gamma = 0.75$: the **West-Choy theoretical model**, in which perfusion scales sub-linearly with mass according to the West-Brown-Enquist metabolic allometric law^{9,35}. This is the lower-bound sensitivity case; it predicts lower flow demand per gram in larger territories relative to the linear null.

So the allometric formulation is a sensitivity generalization: $\gamma = 1.0$ is the primary linear null model, and $\gamma = 0.75$ provides the theoretical lower bound from metabolic allometric scaling. Both are normalized to agree at the reference territory mass $M_{\text{ref},v}$.

16.10 A10. Territory fractions and dominance patterns

The dominance table in the main text is not derived from first principles. It is an anatomical prior. The main mathematical closure requirement is simply that each row sum to one, so that the territorial masses partition the total LV mass:

$$f_{\text{LAD}} + f_{\text{LCx}} + f_{\text{RCA}} = 1.$$

The table satisfies that closure condition in each dominance state: right-dominant $0.42 + 0.30 + 0.28 = 1.00$, left-dominant $0.42 + 0.38 + 0.20 = 1.00$, and co-dominant $0.42 + 0.33 + 0.25 = 1.00$. Holding the LAD fraction approximately constant while redistributing LCx and RCA mass is a modeling assumption motivated by the fact that dominance mainly changes inferior and posterolateral supply rather than anterior LV supply. This is therefore a justified anatomical approximation, not a derivation.

Source distinction: the territory-fraction values by dominance pattern are taken from anatomical cadaveric and imaging studies³³. This source is distinct from³⁴, which validates CT-based LV territorial mass *measurement* against CMR reference (ICC = 0.90) but does not provide the fraction table itself. These two references serve separate roles — anatomical fraction prior and mass-measurement validation — and must not be conflated.

LAD fraction change from prior versions: earlier pre-v14 internal drafts used a LAD fraction of 0.45 for right-dominant cases. The value was updated to 0.42 to match the primary Sakamoto 2013 data more precisely³³; the 0.45 value was a rounded prior from memory rather than from the primary paper. The 0.42 value is now the pre-specified fraction for all four model variants and must not be re-tuned within the development cohort.

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